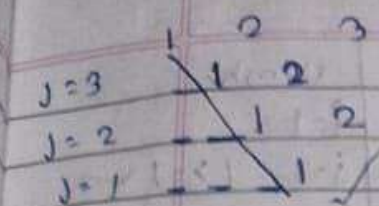


i = n-1 ; ~~i<~~ i>=1 ; i--

Space

s = 1 s< = n-i .

i = 3 s = 4-3 = 1 = 1 2 3

i = 2 s = 4-2 = 2 = 1 2

i = 1 s = 4-1 = 3 = 1

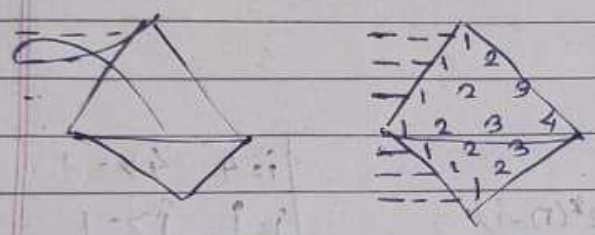
j = 1 j< = i

i = 3 ~~j< = 3~~

s2 j< = 3 1 2 3

i = 2 j< = 2 1 2

i = 1 j< = 1 1



n=4

~~for(int~~

2

1 1

12 21

123 321

1234432 1

1234432

123 321

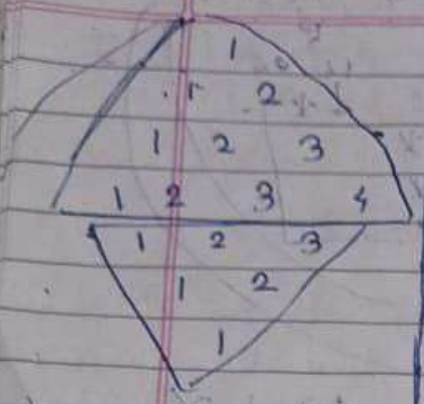
12 21

1 1

```
int n=4
for(int i=1; i<=n; i++){
 for(int j=1; j<=i; j++){
    sop(j);
    }
for(int s=1; s=2*(n-i); s++){
    sop(" ");
    }
for(int j=i; j>=1; j--){
    sop("j");
    }
 sopln();
 }
```

```
for(int i=n; i>=1; i--){
  ~~sop(~~
for(int j=1; j<=i; j++){
    sop(j+" ");
    }
for(int s=1; s<=2*(n-i); s++)
    sop(" ");
    }
for(int j=i; j>=1; j--){
    sop(j+" ");
    }
 }
sopln();
```

Assigment - 3




```
in n=4
for(int i=1; i<=n ; i++){
for(int s=1; s<=n-i; s++){
    sop( " ");
    }
for(int j=1 ; j<=i ; j++){
    sop( j);
    }
    sop(n);
    }
for(int i=n-1 ; n>=1 ; i--){
for(int s=1 ; s<=n-i ; s++){
    sop( " ");
    }
for(int j=1 ; j<=i ; j++){
    sop(j);
    }
    sop(n());
    }
```

```
___ 1   - i=1
__ 1  2  - i=2
_ 1  2  3 - i=3
1  2  3  4 - i=4
_ 1  2  3
__ 1  2
___ 1
```

Pyram

Space s = n - i

s = 4 - 1 = 3 _ _ _

s = 4 - 2 = 2 _ _

s = 4 - 3 = 1 _

s = 4 - 4 = 0

sop();

Number j = 1 , j<= i

sop();

i=1	j=1	1
i=2	j=1	1 2
i=3	j=1	1 2 3
i=4	j=1	1 2 3 4

Q

```
n = 5
for(int i=1; i<=n; i++){
  for(int j=1; j<=n; j++){

    if(i==j || (i+j) = n+1){
        sop(i + " ");
    }
    else{
        sop(" ");
    }
  Sopln();
}
```

$i = 1 \rightarrow 1<=5$

$j = 1 \rightarrow 1<=5$

$i==j$ sop $- j \rightarrow \underline{1}$

$j=2 \rightarrow$ - -

$j=3 \rightarrow$ - -

$j=4 \rightarrow$ - -

$j=5 \rightarrow$ ~~0+1~~ $= \cancel{0+1}\ 6$

$i+j = 6 \rightarrow \underline{1}$

$i=2$

$j=1 \rightarrow$ - -

$j=2 \rightarrow \underline{2}$

$j=3 \rightarrow$ - -

$j=4 \rightarrow$ - - $i+j=6 \rightarrow \underline{2}$

$j=5 \rightarrow$ - -

$i=3$

$j=1 \rightarrow$ - -

$j=2 \rightarrow$ - -

$j=3 \rightarrow \underline{3}$ $\boxed{i+j=6 \,||\, i==j}$

$j=4 \rightarrow$ - -

$j=5$

$i=4$

$j=1 \rightarrow$ - -

$j=2 \rightarrow i+j=6=\underline{4}$

$j=3 \rightarrow$ - -

$j=4 \rightarrow i==j=\underline{4}$

$j=5 \rightarrow$ - -

$i=5$

$j=1 \rightarrow i+j=6 \rightarrow \underline{5}$

$j=2 \rightarrow$ - -

$j=3 \rightarrow$ - -

$j=4 \rightarrow$ - -

$j=5 \rightarrow i=j=\underline{5}$

Page No.
Date

top

n = 3

Q.4 3 3 3 3 3 ← outer - layer - 0
3 2 2 2 3 ← middle layer - 1
3 2 1 2 3
3 2 2 2 3 ← inner layer - 2
left → 3 3 3 3 3 ← right
↑ bottom

n = 3 m = 2 * n - 1
2 * 3 - 1
m = 5 → ~~size~~ 5 * 5.

j = 1 k = size(5)
↓
j = 1 k = size(5)
↓
top = i - 1 = 1 - 1 = 0
bottom = ~~size - i~~
m - i = 4
left = j - 1 = 1 - 1 = 0
right = m - j = 5 - 1 = 4

k = math.min (math.min(top, bottom), Math.
min (left, right)):

math.min(math.min(0, 4) , math min(
0, 4)
= math.min(0, 0).
k = 0

int value = n - k
= 3 - 0
value = 3
print (value) = 3

(4,1) (4,2) (4,3) (4,4) (4,5)
3 3 2 2 3

reflecte (5,1) (5,2) (5,3) (5,4) (5,5)
3 3 3 3 1

num = ~~1 2 3 4 5 6 7 8 9 10 11 12 13 14 15~~

Q5

```
        1      -i=1
      2   3    -2
    4   5   6  -3
  7   8   9  10 -4
-11  12  13   -3
  -14  15    -2
    -16      -1
```

i=1
s=1 1k=4-1=3 —
j=1 1k=1 1
n+1
i=2 s=1 1k=2 —
j=1 1k=2 2
n+1
2k=2 3
n+1
i=3 s=1 1k=1 —
j=1 1k=3 4
n+1
2k=3 5
n+1
3k=3 6
n+1

i=4
j=1 1k=4 7
n+1
2k=4 8
n+1
3k=4 9
n+1
4k=4 10
n+1

i=n-1
4-1=3
i=3
s=1 1k=1 —
j=1 1k=3 11
n+1 2
2k=3 12
3k=3 13
n+1

i=2
s=1 1k=2 --
2k=2
j=1 1k=2 14
2k=2 15
n+1

i=1
s=1 1k=3 ---
j=1 1k=1 16
n+1

$i = 1$

top = i-1

$\rightarrow (1,1)$ $(1,2)$ $(1,3)$ $(1,4)$ $(1,5)$

3 3 3 3

~~i = 2~~

~~j = 1~~

$i = 2$ — top = i-1 = 2-1 = 1

$j = 1$ bottom = m-i = 5-2 = 3] — 1

left j-1 = 0 → 0] — 0

(2,1) (2,2) (2,3) (2,4) (2,5)

3 2 2 2 3

n-k = 8-0 = 3

$i = 2$

$j = 2$ → left — j-1 = 2-1 = 1] — top, bottom 1 → 1] → 1

right = 5-2 = 3

n-k = 3-1 = 2

$i = 3$ top = i-1 = ~~3-1 = 2~~ 3-1 = 2] — 2

$j = 1$ bottom = 5-3 = 2] — 0

right = 5-1 = 4

left = j-1 = 0

k = 0 n-k = 3-0 = 3

(3,1) (3,2) (3,3) (3,4) (3,5)

3 2 1 2 2

$i = 3$

$j = 2$ left = j-1 = 2-1 = 1] — 1 → 1] — 1

right = 5-2 = 3

n-k = 3-1 = 2

$i = 3$ left = 3-1 = 2] — 2

$j = 3$ right 5-3 = 2] — 2

Q6

1 ← i=1

0 1 ← 2

1 0 1 ← 3

0 1 0 1 ← 4

1 0 1 ← 3

0 1 ← 2

1 ← 1

j=1 k=4

S=1 k=3 — — —

2<=3

3<=3

4<=4

j=1 k=

j=1 1>=1

└→ j%2

1%2=1

i=2 2<=4

S=1 k=2 — —

2<=2

j=2 2>=1

└ 2%2=0

1>=1

└ 1

i=3

S=1 k=3 —

j=3 3>=1

└ 1

2>=1

└ 0

1>=1

└ 1

i=4

S=1 k=n-i=0

j=4 4>=1

└ 0

3>=1

└ 1

2>=1

└ 0

1>=1

└ 1

j=3 3>=1

S=1 k=4-3=1

j=3 3>=1

└ 3%2=1

2>=1

└ 0

1>=1

└ 1

i=2 2>=1

S=1 k=4-2=2 — —

j=2 2>=1

└ 0

1>=1

└ 1

j=1 1>=1

S=1 k=3 — — —

j=1 1>=1

└ 1

1

0 1

1 0 1

0 1 0 1

1 0 1

0 1

1